

École Nationale des Sciences Appliquées de Berrechid

Génie Mécatronique et Systèmes Intelligents

End-of-Year Internship Project

Final internship report

Integrated Brake-by-Wire and Steer-by-Wire Control Allocation

A fault-tolerant vehicle simulation demonstrator

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ENSA Berrechid – SEGULA Technologies internship framework

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Dedication

To our families, for their encouragement throughout our studies.

To our teachers, whose work has helped us build the scientific and technical foundations of this project.

To our friends and fellow students, for the discussions, mutual support and shared desire to learn.

Laaraki Mohamed Amine
Aymen Bentaleb

Acknowledgements

We thank SEGULA Technologies and ENSA Berrechid for the academic and professional framework in which this internship project was developed. Working on integrated braking and steering control offered an opportunity to connect mechatronics, vehicle dynamics, numerical optimization and engineering verification.

We express our gratitude to Mr Elmehdi Chokri, project guarantor, for his guidance and support throughout our work on this simulation demonstrator.

We warmly thank our academic supervisors, Pr Said Ben Alla and Pr Issam Amellal, and the teaching staff of ENSA Berrechid for their role in our training and the academic supervision of this work.

Our thanks also go to those who will evaluate this report, and to everyone who supported us during the project. This report is an opportunity to share both the engineering solution developed and the lessons learned from its implementation.

Abstract

In this end-of-year internship project, we develop a functional simulation demonstrator for integrated Brake-by-Wire and Steer-by-Wire control allocation. Four wheel brakes and a shared steering rack driven by two independent functional channels are coordinated by a constrained quadratic program. The controller redistributes longitudinal force and yaw-moment requests according to diagnosed actuator availability, while a nonlinear 13-state vehicle model represents tire saturation, combined force capacity, load transfer and actuator dynamics. Physically unattainable demands are treated through bounded best-effort allocation.

The completed demonstrator is supported by 45 MATLAB unit tests, five nominal maneuvers and 30 original fault scenarios. Differential-Braking Backup Steering (DBBS) satisfies the project pulse criteria in all eight assessed cases: worst path deviation is 0.295 m, worst yaw-rate RMSE is 0.0115 rad/s and maximum deceleration is 0.277 g. In the evaluated front-brake-loss authority maneuvers, at least 61.1% of the nominal upper-bound braking capability is retained, with peak sideslip limited to 0.0605 rad. Targeted 1 ms refinement confirms the principal results.

The final academic scope is a reproducible, model-based demonstration at a documented reference operating condition, complemented by a qualitative safety analysis. Broader operating-envelope guarantees, abrupt faults during fully saturated braking and hardware validation are organized as future perspectives. The numerical records are preserved, and completion of the academic demonstrator is distinguished from production certification or an all-condition safety claim.

Keywords: BbW, SbW, DBBS, control allocation, quadratic programming, vehicle dynamics, fault tolerance, MATLAB.

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Abbreviations and notation

BbW / SbW	Brake-by-Wire / Steer-by-Wire
DBBS	Differential-Braking Backup Steering
QP	Quadratic program
RMSE	Root mean square error
FMEA / AMDEC	Failure modes and effects analysis
FTA / DFA	Fault-tree / dependent-failure analysis
ASIL / ECU	Automotive Safety Integrity Level / Electronic Control Unit
CDC / PFA	Project specification / end-of-year project
F_x, F_y, F_z	Longitudinal, lateral and normal forces (N)
M_z, r, β	Yaw moment (N m), yaw rate (rad/s), sideslip (rad)
δ, μ	Road-wheel/rack angle (rad), friction coefficient
h_p, h_k	Physical and diagnosed health masks

General introduction

Electrically commanded braking and steering create a coordination problem as well as an actuator-control problem. When one effector becomes unavailable, a command that was feasible with the complete actuator set may no longer be attainable. A useful degraded response therefore requires explicit prioritization, physical limits and a method for exploiting the effectors that remain available.

We address this problem by treating the brake and steering subsystems as one over-actuated plant. Steering and differential braking can both influence yaw, but they are not interchangeable: braking reduces speed and shares the tire friction budget with lateral force. This coupling motivates a nonlinear vehicle model and a constrained allocation strategy.

We follow a progressive engineering sequence: formalize the functional architecture, verify the plant without allocation, connect nominal control, introduce diagnosed channel losses, improve degraded behavior, then preserve and independently audit the results. The deliverable is a working simulation system accompanied by a reproducible evidence set and this bilingual report.

In the following chapters, we present the host organization and project objectives, the literature review, the functional architecture, the mathematical model, the allocator, our simulation results, the safety rationale, and future perspectives. We evaluate our demonstrator within the operating conditions and scenario set defined in Chapter 1; broader industrial validation remains a subsequent development stage.

Host organization and project framing

1.1 SEGULA Technologies and the Moroccan context



Figure 1.1 – SEGULA Technologies, host engineering context.

SEGULA Technologies is an engineering group. Its automotive activities include development, testing and technology consulting for vehicle manufacturers and suppliers. The Moroccan organization presents automotive engineering through its Casablanca design office and aerospace activities through SIMRA. Its official website also lists locations in Agadir and Tanger. These elements situate the internship within an engineering and industrial-development environment [12, 13].

Our project concerns control engineering and numerical demonstration. It does not claim integration into a named customer's production vehicle, access to proprietary vehicle data or validation by an industrial vehicle program. The reference vehicle is assembled from published and explicitly assumed parameters.

1.2 Academic organization

Table 1.1 – Project identification.

Institution	ENSA Berrechid
Programme	Génie Mécatronique et Systèmes Intelligents
Prepared by	Laaraki Mohamed Amine; Aymen Bentaleb
Academic supervisors	Pr Said Ben Alla; Pr Issam Amellal
Project guarantor	Elmehdi Chokri
Reference environment	MATLAB R2026a Update 5; Optimization Toolbox
Deliverable	Simulation demonstrator and FR/EN report

1.3 Project objectives and scope

Our objective is to develop and document a functional simulation demonstrator for integrated braking and steering control. We use the CDC to guide our architecture and quantitative

benchmarks, and assess the system on a reference vehicle model and a defined set of nominal and degraded maneuvers. Our work covers modeling, constrained allocation, selected actuator losses and analysis of residual performance. The table below distinguishes the capabilities we demonstrate from the developments we identify as future perspectives.

Table 1.2 – Completed deliverable and future development boundary.

Current deliverable	Future perspectives
Nonlinear reference vehicle; common rack and four brakes	Vehicle-specific identification and experimental correlation
Nominal allocation; selected fail-silent channel losses	Stuck actuators, sensor/ECU/common-cause faults
Measured braking and DBBS performance in the catalog	Abrupt faults during fully saturated braking; continuous recovery/envelope guarantees
Traceable simulation and qualitative safety rationale	Hardware independence, real-time timing, certification

Appendix C gives the clause-by-clause status of the engineering baseline, so that demonstrated results are never read as blanket compliance. We preserve the simulation records so that each result remains traceable to its test conditions and controller version. Our additional investigations of saturated transitions inform the future-work chapter. Throughout this report, we explain the demonstrated behavior, our design choices and the practical next steps for the system.

1.4 Method and project progression

Table 1.3 – Engineering work sequence.

Stage	Question	Produced evidence
1	What is redundant?	Architecture and fault semantics
2	Does the plant behave consistently?	Open-loop and actuator benches
3	Can demands be allocated nominally?	Five time-varying maneuvers
4	What remains after a channel loss?	30 scenarios and residual authority
5	How can degraded behavior improve?	Rear reserve and DBBS feedback tuning
6	Can the result be explained and reproduced?	Independent audits, source snapshots, final report

We carried out most of this work together, from the literature review and design to result analysis and report writing. Our contributions are joint.

Literature review and design choices

We examine how redundant vehicle actuators can preserve braking and lateral control after a fault. Our review connects three questions: which physical mechanisms provide backup authority, how that authority can be allocated under constraints, and which assumptions are necessary before claiming fault tolerance. The references identified during Gate 1 inform our design; their vehicle architectures and test conditions are distinguished from our own.

2.1 Hardware redundancy and cross-domain compensation

Duplicating an actuator preserves a familiar control mechanism, provided that the surviving branch has sufficient capability and is not defeated by the same cause. Cross-domain compensation instead uses another vehicle function to create a useful force or moment. The two approaches can coexist: our steering drives provide local redundancy, while differential braking provides a second means of influencing yaw.

Eckstein’s account of the SpeedE research vehicle describes wheel-individual steering and cross-domain compensation. A faulty steering intervention can be counteracted by the other steered wheel together with rear-axle torque vectoring [2]. The important precedent is reasoning at vehicle-function level rather than equating redundancy with identical duplicated components.

We retain this functional principle, but our plant has one common rack and no independently commanded propulsion torques. It cannot counter-steer one front wheel against the other. Its available redundancy must therefore be assessed using its own four brake-force channels and shared steering angle. Results obtained with wheel-individual steering cannot be transferred directly to this architecture.

2.2 Braking as a backup for steering

Two different mechanisms appear under the broad expression “steering by braking”. Dominguez-Garcia, Kassakian and Schindall propose brake-actuated steering in which asymmetric wheel forces actuate the steering mechanism after mechanical decoupling from the failed primary actuator [3]. This approach depends on steering geometry and the mechanical freedom of the rack. It requires a model of steering torque and motion.

Kissai, Mouton and Monsuez study EPAS backup through differential braking: separate wheel braking produces a vehicle yaw moment, with allocation coordinating the available brakes [4]. Their weighted least-squares formulation provides a useful precedent for bounded allocation. Its relevance to our work is the explicit management of brake authority, rather than an assumption that braking recreates every capability of the steering system.

We adopt differential-braking backup steering (DBBS) in this second sense. When both steer-

ing drives are lost, the model holds the existing rack angle and the allocator creates additional yaw through brake asymmetry. We do not model brake-induced rack motion. This distinction fixes what our simulations can establish: a controlled lateral response with deceleration, under a specified rack state and tire-force budget.

2.3 Steering assistance after brake loss

Jiang and colleagues address single-wheel electromechanical-brake failure through coordinated brake-torque redistribution and active front steering [5]. Their treatment of the conflict between deceleration and lateral stability is directly relevant to our front-brake-loss cases. Brake utilization and lateral support cannot be selected independently when both consume the same tire adhesion.

Our response is to retain a larger rear lateral-force reserve after diagnosed front-brake loss. This is a project-specific control choice evaluated with our nonlinear plant, not a reproduction of the published controller or its numerical results. The comparison explains why a small reduction in braking utilization can be justified when it substantially reduces sideslip.

Han and colleagues investigate steering-system failure in four-wheel independent by-wire steering electric vehicles [6]. Gate 1 identified their hierarchical predictive-control and torque-distribution approach as another example of actuator coordination. We classify it here as related steering-fault work, not evidence that steering alone restores lost braking. Its additional steering and drive degrees of freedom differ from our common-rack, braking-only allocation.

2.4 Control allocation methods

The survey by Johansen and Fossen separates the motion-control demand, its allocation among effectors, and local actuator control [1]. This hierarchy is useful for our implementation: generalized force and yaw moment remain the interface between vehicle-level objectives and actuator-level constraints. The following comparison explains our selection; it is not an experimental ranking of algorithms.

Table 2.1 – Allocation alternatives and implications for our design.

Method	Useful property	Design consideration
Pseudoinverse	Direct solution of a linear force mapping	Clipping alone can leave avoidable tracking error; redistribution requires added logic
Daisy chain	Explicit priority between actuator groups	The group order determines how remaining authority is used
Bounded WLS / QP	Weighted error and actuator bounds in one problem	Requires scaling, numerical checks and defined fallback behavior

We select a quadratic program with a locally affine force mapping, bounds and command-rate limits. Force and yaw errors are scaled before weighting so that units do not set the priority implicitly. When a demand exceeds available authority, minimizing a bounded residual

preserves useful actuation. The nonlinear plant remains the reference for actual delivered force and moment; minimizing the affine allocation error is not itself a nonlinear stability proof.

2.5 Positioning of our contribution

Our contribution is the integration and verification of these principles in one reproducible academic demonstrator. We combine explicit fault masks, independent physical and diagnosed fault timing, a two-drive common-rack model, constrained best-effort allocation and nonlinear evaluation. We then use the observed braking/sideslip trade-off and DBBS path error to improve the degraded controller.

The reviewed work supports the choice of mechanisms and methods, but does not establish our numerical performance. Our own results are presented in Chapter 6. Likewise, ASIL D motivates the safety analysis without turning simulated redundancy into demonstrated hardware independence. The FMEA and dependency review connect our tested fault consequences to the additional work required for a physical system.

Functional architecture and control objectives

3.1 Integrated actuation chain

The control chain comprises a maneuver reference, a yaw-rate servo, a QP allocator, actuator dynamics and the nonlinear plant. Plant state feedback updates allocation at each 2 ms sample. The plant uses the physical fault mask; allocation uses the delivered diagnosis mask. This separation exposes the period during which a failed actuator is still believed available.

$$u = [F_{x,FL}, F_{x,FR}, F_{x,RL}, F_{x,RR}, \delta_c]^T, \quad h = [h_{FL}, h_{FR}, h_{RL}, h_{RR}, h_A, h_B]^T.$$

There is one rack angle, not two angles to sum. Two independently masked inner velocity loops contribute to the same rack motion. The healthy pair is limited to 400 degrees/s; each drive is assigned 240 degrees/s and the full angle range. These are functional design assumptions rather than demonstrated torque capacity.

Fault	Implemented physical effect
One brake	Contact force removed immediately. Latent actuator state decays but cannot generate tire force.
One steering drive	Its rack-velocity contribution is removed. The survivor retains its former share until diagnosis.
Both drives	Rack holds its current angle. Brakes generate backup yaw.
Power / communication	Equivalent to a mask only if the real effect is local, isolated and fail-silent.

A held rack does not represent a floating rack, arbitrary jam or opposing motor torque. Lost/corrupt driver requests, sensor failures and common ECU faults also require different models. Both steering drives unavailable can be two independent losses or one shared-path failure: the physical architecture must decide. Six software flags alone do not demonstrate six independent hardware domains.

3.2 Control objectives and fault timing

The requested body acceleration and yaw response are translated into generalized force and moment. The allocator does not directly command a vehicle path; it tracks these quantities using the available actuators. DBBS means yaw generation through left/right braking asymmetry. It

does not refer to steering the rack through scrub-radius effects.

$$F_{x,d} = m(a_{x,ref} - v_y r), \quad M_{z,d} = I_z \left(\dot{r}_{ref} + \frac{r_{ref} - r}{\tau_r} \right).$$

Physical loss occurs at t_p . The diagnosis becomes available at $t_f = t_p + t_d$, sampled on the controller grid. Using separate masks prevents the controller from reacting to a future fault. Diagnosis delays of 0, 10 and 50 ms are imposed as sensitivity cases, not measured detector performance.

3.3 Benchmarks used in the demonstrator

Table 3.1 – Scenario evaluation benchmarks.

Nominal tracking	$\text{RMSE}(F_x) \leq 250 \text{ N}; \text{RMSE}(r) \leq 0.012 \text{ rad/s}$
DBBS yaw pulse	$ r_{ref} \leq 0.15 \text{ rad/s}; \text{RMSE}(r) \leq 0.03 \text{ rad/s}$
DBBS path / deceleration	$ e_{lat} _{\max} \leq 0.50 \text{ m}; a_{brake} \leq 0.35g$
Residual braking plateau	$a_{brake} \geq 0.60\mu g; M_z \leq 567.137 \text{ N m}$
Operating-domain screen	$v_x > 0.5 \text{ m/s}; \beta \leq 0.15 \text{ rad}$
Mask / recovery diagnostics	10 ms; 10% band, 50 ms entry and 100 ms dwell

The braking yaw limit is 5% of a feasible nominal yaw witness and is conservative for this benchmark. Recovery uses an affine reference and is reported as a diagnostic. The final academic scope does not promote these sampled tests into an all-condition physical guarantee.

Table 3.2 – Link between engineering questions and report evidence.

Question	Evidence	Interpretation
Are commands bounded?	QP bounds, rate logs and unit tests	Nominal and catalog checks; saturated transitions discussed separately
Does allocation react to loss?	Separate physical/known masks and saved timing traces	Imposed diagnosis delay, not detector qualification
What authority remains?	Braking plateaus, DBBS pulses and sampled witnesses	Quantified at stated states and maneuvers
What does redundancy cover?	Full FMEA and dependency register	Functional rationale; hardware independence requires further evidence

3.4 Implemented control chain

Figure 3.1 reconstructs the functional structure of the executed MATLAB scripts. Each block corresponds to a named function or a model stage. It is an implementation diagram, not a screenshot of the historical Simulink prototype. Solid arrows carry control or state signals; dashed arrows carry imposed health information.

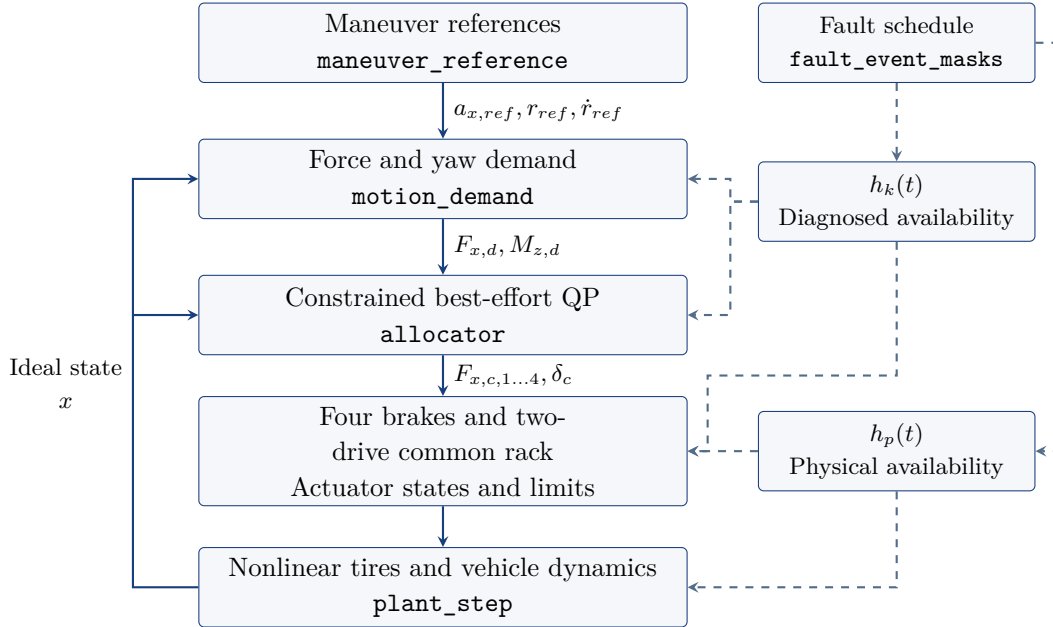
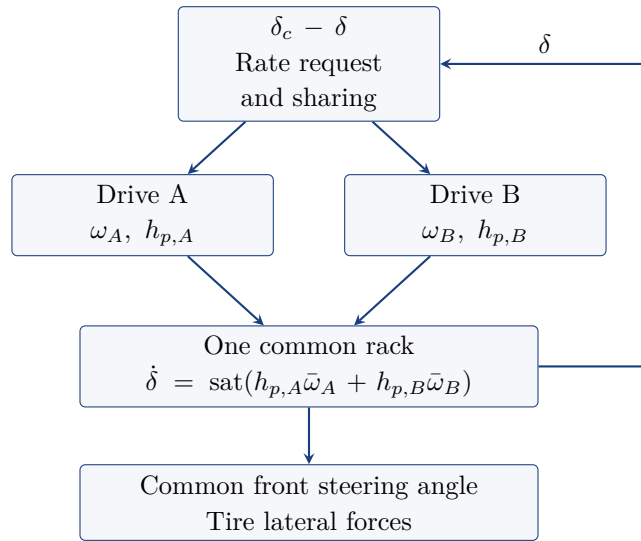


Figure 3.1 – Executed control and fault-injection chain. The controller also retains the previous command to impose rate limits.

The nominal controller period is 2 ms. The physical mask removes actuator capability at fault onset; the known mask changes only after the imposed diagnosis delay. The state feedback comes directly from the simulated plant. This architecture isolates diagnosis-latency sensitivity without claiming that a sensor-based detector or state estimator has been implemented.

3.5 Common-rack actuation and available redundancy



Two healthy drives: $400^\circ/\text{s}$ aggregate limit One healthy drive: $240^\circ/\text{s}$
 Both lost: held rack; brake asymmetry supplies backup yaw

Figure 3.2 – Functional steering model with two velocity-loop contributions and one mechanical degree of freedom.

Each motor loop has a 10 ms time constant; the outer angle-to-rate response uses 20 ms. The surviving drive receives the remaining request after diagnosis. Its assumed rate capability is 60% of the healthy pair's limit, while the allowed angle remains $\pm 35^\circ$. These are functional model parameters, not a measured torque-under-load specification. The four brake channels remain separately commanded. Their longitudinal tire forces contribute both to deceleration and to yaw according to wheel position. All channels share vehicle load transfer and tire adhesion. Thus the drawing of separate actuator channels does not imply independent power supplies, buses, sensors or safety controllers. Those dependencies are examined in the safety chapter and Appendix B.

Chapter 4

Vehicle and actuator modeling

4.1 State variables and equations of motion

The planar state is

$$x = [v_x, v_y, r, X, Y, \psi, \delta, F_{x,FL}, F_{x,FR}, F_{x,RL}, F_{x,RR}, \omega_A, \omega_B]^T.$$

For body-frame forces and yaw moment, the motion equations are

$$\begin{aligned} \dot{v}_x &= F_x/m + v_y r, & \dot{v}_y &= F_y/m - v_x r, & \dot{r} &= M_z/I_z, \\ \dot{X} &= v_x \cos \psi - v_y \sin \psi, & \dot{Y} &= v_x \sin \psi + v_y \cos \psi, & \dot{\psi} &= r. \end{aligned}$$

Front and rear slip angles are

$$\alpha_f = \delta - \text{atan2}(v_y + l_f r, v_x), \quad \alpha_r = -\text{atan2}(v_y - l_r r, v_x).$$

A simplified Magic Formula preserves the sourced small-slip axle stiffness. Each wheel's remaining lateral capacity is $\sqrt{\max((\mu F_{z,i})^2 - F_{x,i}^2, 0)}$. The axle peak is the sum of these capacities, and lateral force is distributed by capacity share. Front forces are rotated into the body frame before summing force and moment.

$$F_y = D \sin[C \arctan\{B\alpha - E(B\alpha - \arctan B\alpha)\}], \quad B = \frac{C_\alpha}{CD}.$$

Quasi-static load transfer uses an estimated longitudinal acceleration from actuator force and $a_y \simeq v_x r$. Negative raw wheel loads trigger a check even though force evaluation clips them to zero. Brake states use a 20 ms lag limited to 40 kN/s. A 20 ms rack outer loop distributes requested velocity to two 10 ms motor loops. Classical RK4 integrates between allocation samples. Validity is restricted to $v_x > 0.5$ m/s and $|\beta| \leq 0.15$ rad, with $\beta = \text{atan2}(v_y, v_x)$. The model omits wheel-speed dynamics, slip-ratio control, rack inertia, road-load torque, suspension transients and sensor dynamics. Passing numerical tests is model verification, not experimental validation of a real vehicle.

4.2 Load transfer and combined-force coupling

Longitudinal load transfer changes the front/rear distribution according to the estimated acceleration. Lateral transfer redistributes each axle's load between its left and right wheels. The model evaluates raw loads before clipping, allowing lift-off conditions to be detected numerically.

$$\Delta F_{z,x} = \frac{mh_{CG}a_x}{L}, \quad F_{zf} = mg \frac{l_r}{L} - \Delta F_{z,x}, \quad F_{zr} = mg \frac{l_f}{L} + \Delta F_{z,x}.$$

For each wheel, longitudinal and lateral forces share the same friction-circle budget. Consequently, using nearly all available tire force for braking leaves little lateral support. This is central to the degraded-control design: yaw balance alone is not sufficient to preserve sideslip behavior.

4.3 Parameter provenance

REF-2026-02 is a literature-based reference vehicle, not an identified target vehicle. S denotes sourced, C calculated, A assumed and R derived from the project requirement interpretation. Mixed provenance is intentional and must accompany the results.

Parameter	Value	Provenance
m, I_z	1590 kg; 2059.2 kg m ²	S, Wang et al. [7]
l_f, l_r	1.05 m; 1.61 m	S [7]
L, w_f	2.66 m; 0.6053	C: $l_f + l_r, l_r/L$
t_w	1.53 m	A: width used as track surrogate [8]
h_{CG}	0.637 m	S surrogate [8]
g, μ	9.80665 m/s ² ; 0.90	S [9]; A
$C_f = C_r$	66000 N/rad	C: 2×33000 [7]
C, E	1.9; 0.97	A
D, B	Capacity-dependent	C, previous section
$\delta_{\max}$	35°	A
Rack / drive rate	400°/s; 240°/s	R/A; A
Brake force rate	40000 N/s	R/A
$\tau_{Fx}, \tau_\delta, \tau_\omega$	20, 20, 10 ms	A
Yaw servo	0.25 s / 0.15 s (DBBS)	A
Controller step	2 ms	R
Initial speed	60 km/h	R
Force reserve	2% / 8% rear after front loss	A
Objective scales	5000 N; 1000 N m	A
Effort weight / trust angle	10^{-6} ; 0.5°	A

The 1.53 m value in [8] is vehicle width, not a measured track. Transferring it into the track parameter is explicitly an assumption. Geometry, tire shape, friction and actuator dynamics therefore limit representativeness. The 60% surviving steering rate is built into the functional model; the bench checks implementation consistency rather than discovering a measured capacity. One unsaturated 20 ms lag reaches 91.8% in 50 ms. This calculation does not prove the cascaded steering recovery requirement. Measured vehicle data and parameter sweeps would be needed before transferring results to a particular vehicle.

4.4 Verification before allocation

Plant verification preceded closed-loop allocation. Straight braking checks longitudinal deceleration and near-zero yaw under symmetric commands. Step steering checks the sign and qualitative transient of yaw and lateral response. High-force cases check tire saturation and positive loads. Separate common-rack benches exercise two healthy drives, each single-drive loss and complete drive loss.

The preserved Gate 2 campaign includes three plant-only cases and four steering cases. This hierarchy is important: a nominal closed-loop trajectory alone would not reveal whether an incorrect vehicle equation had been compensated accidentally by the controller. The tests establish internal model consistency at the selected conditions; physical vehicle correlation is a subsequent perspective.

Allocation strategy and degraded-mode design

5.1 Constrained quadratic allocation

At each sample, the allocator approximates generalized output by $\hat{y} = c + Bu$, with desired $y_d = [F_{x,d}, M_{z,d}]^T$. The offset matches actual nonlinear forces at the current anchor. Brake columns use force geometry with frozen lateral forces and loads. A centered secant estimates steering yaw sensitivity. Steering is not credited with a longitudinal sensitivity in the QP, avoiding an identified saturation-induced steering/braking feedback problem. Actual longitudinal steering effects remain in the offset and plant.

$$\min_z \frac{1}{2} \|W(c + BSz - y_d)\|_2^2 + \frac{\rho}{2} \|z\|_2^2, \quad u = Sz,$$

$$W = \text{diag}(1/5000, 1/1000), \quad \rho = 10^{-6}.$$

S scales the four forces and rack angle into dimensionless decisions. The strictly positive effort penalty regularizes the Hessian. The constraints intersect tire/actuator bounds, healthy command-rate bounds, fault masks and a local steering trust interval. Brake forces remain non-positive. Nominal integrated commands retain a 2% longitudinal reserve; the rear-brake policy becomes 8% after a diagnosed front-brake loss with both rear brakes healthy.

$$\begin{aligned} -\gamma_i \mu F_{z,i} \leq u_i \leq 0, & \quad |u_i - u_{i,\text{prev}}| \leq \dot{F}_{\text{max}} \Delta t, \\ |\delta_c| \leq 35^\circ, & \quad |\delta_c - \delta_{c,\text{prev}}| \leq \dot{\delta}_{\text{avail}} \Delta t. \end{aligned}$$

A failed brake has zero force authority. With both steering drives lost, the rack command equals the held actual angle. If physical bounds and rate bounds have an empty intersection, physical bounds take priority and the healthy-channel override is explicitly logged as a validation issue. No such healthy override occurs in the cited accepted numerical runs. Demand infeasibility is different from solver failure. An infeasible request still yields the closest weighted bounded output. A failed QP instead clips the previous command into the current bounds and raises a warning. This fallback is not an independent safety controller. The finite-dimensional QP prediction is also not a global nonlinear reachable-set oracle.

5.2 Rear lateral reserve after front-brake loss

The final controller sets $\gamma_i = 0.98$ in normal allocation. When exactly one front brake is diagnosed unavailable and both rear brakes remain healthy, rear channels use $\gamma_i = 0.92$. This leaves a larger theoretical lateral-force budget at the commanded longitudinal limit. The trigger

is diagnosed health, never a test-case identifier.

$$\sqrt{1 - 0.98^2} = 0.199, \quad \sqrt{1 - 0.92^2} = 0.392.$$

The reserve is a control choice, not a change to road friction or the tire equations. A design sweep over six rear-utilization values exposed the balance between residual braking and sideslip. The selected value retains more than 60% braking in the assessed plateau while substantially reducing peak sideslip. The margin is quantified in Chapter 6 rather than assumed to apply to all vehicles.

When a larger reserve is introduced, the controller approaches it at the permitted release rate. The reserve is softer than the physical bound. A genuinely empty intersection of physical and rate constraints is retained as a separate transition-model issue, discussed in the future-work chapter.

5.3 DBBS yaw-feedback scheduling

$$\tau_r = \begin{cases} 0.15 \text{ s}, & h_{k,A} = h_{k,B} = 0, \\ 0.25 \text{ s}, & \text{otherwise.} \end{cases}$$

Faster yaw feedback after complete diagnosed steering-drive loss improves pulse tracking and the resulting equal-distance path comparison. Single-drive loss and nominal operation retain the original gain. The external yaw reference is unchanged; this is not a hidden relaxation of the maneuver. A speed-scaled reference alternative was explored but not selected because it increased the measured path deviation.

5.4 Implementation and verification discipline

The controller is divided into reference generation, demand construction, constrained allocation and plant integration. Tests check default/explicit parameters, fault-mask behavior, rear-reserve activation, rate-compatible reserve introduction and diagnosis-based feedback scheduling. The final reference suite contains 45 passing MATLAB unit tests. A local Python audit adds eight tests for saved-data timing arithmetic. These are distinct suites and are not counted as 53 MATLAB tests.

Simulation campaign and achieved performance

6.1 Execution conditions and evidence

The reference campaign uses MATLAB Online R2026a Update 5 with Optimization Toolbox, $\Delta t = 2$ ms, $v_x(0) = 60$ km/h and $\mu = 0.90$. The final physical parameter set is REF-2026-02 and controller revision ALLOC-2026-04. Actual forces and moments are obtained from the nonlinear plant, not from the QP prediction. All figures below are plotted from preserved MATLAB CSV data; local plotting does not rerun the simulation.

6.2 Nominal maneuvers

G2-BRK is progressive straight braking at $0.20g$. G2-STR follows the linear bicycle yaw response associated with a 1° road-wheel input. G2-CMB combines $0.12g$ braking with a 0.06 rad/s yaw demand. G2-YAW applies a windowed sinusoidal reference over 5 s, with coefficient 0.08 rad/s. G2-SAT requests $1.2\mu g$ and then releases the brakes. The four tracking tests last 6 s; the capacity test lasts 4 s.

Pulse ramps use $w(s) = 3s^2 - 2s^3$ on $0 \leq s \leq 1$, with zero endpoint slope and an analytic derivative for the yaw demand. The sinusoid uses a $\sin^2$ window. These references are reproducible project maneuvers, not standardized vehicle tests.

Table 6.1 – Nominal maneuver results.

Case	RMSE(F_x), N	RMSE(r), rad/s	$ \beta _{max}$, rad
G2-BRK	44.65	0.00000	0.00000
G2-STR	20.62	0.00213	0.00511
G2-CMB	28.54	0.00179	0.00267
G2-YAW	6.27	0.00172	0.00326
G2-SAT	1541.32	0.00000	0.00000

The four tracking maneuvers meet their force and yaw RMSE criteria. The saturation maneuver deliberately requests more than the physical capacity, so it is evaluated on delivered capacity, bounded response and release behavior instead of force tracking. It reaches a saturated braking ratio of 0.98. The five saved nominal result summaries match the prior verified reference within the local audit tolerance.

6.3 Fault-scenario catalog

Table 6.2 – Thirty original fault scenarios.

N	Condition	Timing
18	FL/FR/RL/RR or steering A/B loss during combined maneuver	Onset 2 s; diagnosis 0/10/50 ms
6	Complete steering-drive loss, positive DBBS pulse	Onset 1 or 3 s; diagnosis 0/10/50 ms
4	Single brake loss during progressive braking	Onset 2 s; immediate diagnosis
2	Complete steering-drive loss, negative DBBS pulse	Onset 1 or 3 s; diagnosis 10 ms

The 22 single-channel cases and eight complete-drive-loss backup cases are kept distinct. Four matched nominal references support comparison. Complete steering loss holds the actual rack position, giving centered and turning-rack cases. The pulse ramps to ± 0.15 rad/s, holds, then returns smoothly to zero; it is a project maneuver, not a standardized lane-change test.

6.4 Measurement conventions

Lateral deviation is measured in the nominal heading frame at equal cumulative travelled distance. This avoids confusing a speed difference with a spatial tracking error. The nominal record includes sufficient coasting distance to cover the faulted path without extrapolation. With nonzero sideslip, this is a heading-normal deviation, not the exact shortest geometric distance.

$$e_{lat}(s) = -(X_f(s) - X_n(s)) \sin \psi_n(s) + (Y_f(s) - Y_n(s)) \cos \psi_n(s).$$

Yaw RMSE and DBBS peak deceleration are evaluated after physical onset. The first 200 ms yaw-continuity diagnostic compares synchronized nominal and faulted signals. These definitions are retained across controller revisions so that improvements are attributable to the design rather than the measurement method.

6.5 DBBS performance

Table 6.3 – DBBS results with ALLOC-2026-04.

Case	$ e_{lat} _{max}, \text{ m}$	RMSE(r)	a_{max}/g
DBBS-T1-D00	0.2945	0.01146	0.2771
DBBS-T3-D00	0.2617	0.00830	0.2328
DBBS-T1-D10	0.2945	0.01146	0.2771
DBBS-T3-D10	0.2617	0.00830	0.2328
DBBS-T1-D50	0.2945	0.01146	0.2771
DBBS-T3-D50	0.2616	0.00830	0.2328
DBBS-NEG-T1-D10	0.2945	0.01146	0.2771
DBBS-NEG-T3-D10	0.2617	0.00830	0.2328

We observe negligible sensitivity to the tested 0/10/50 ms diagnosis delays because diagnosis precedes the pulse in the 1 s fault cases: even the latest flag arrives at 1.05 s, before the pulse starts at 1.5 s. The 3 s cases occur on a constant-reference plateau with the rack held at its existing angle. This finding is specific to these fault timings and the held-rack model; it does not establish general latency robustness during rapid transients. By contrast, Table 8.1 examines abrupt brake loss during saturated braking, which immediately disturbs the force balance.

All eight cases satisfy the three maneuver criteria. The worst path deviation is reduced from 0.5086 m in the initial design to 0.2945 m in the corrected design, approximately a 42% reduction. The maximum deceleration increases from 0.2577 g to 0.2771 g, illustrating the physical cost of better yaw tracking through differential braking. The comparison is between two revisions of the same architecture, not between unrelated controller families.

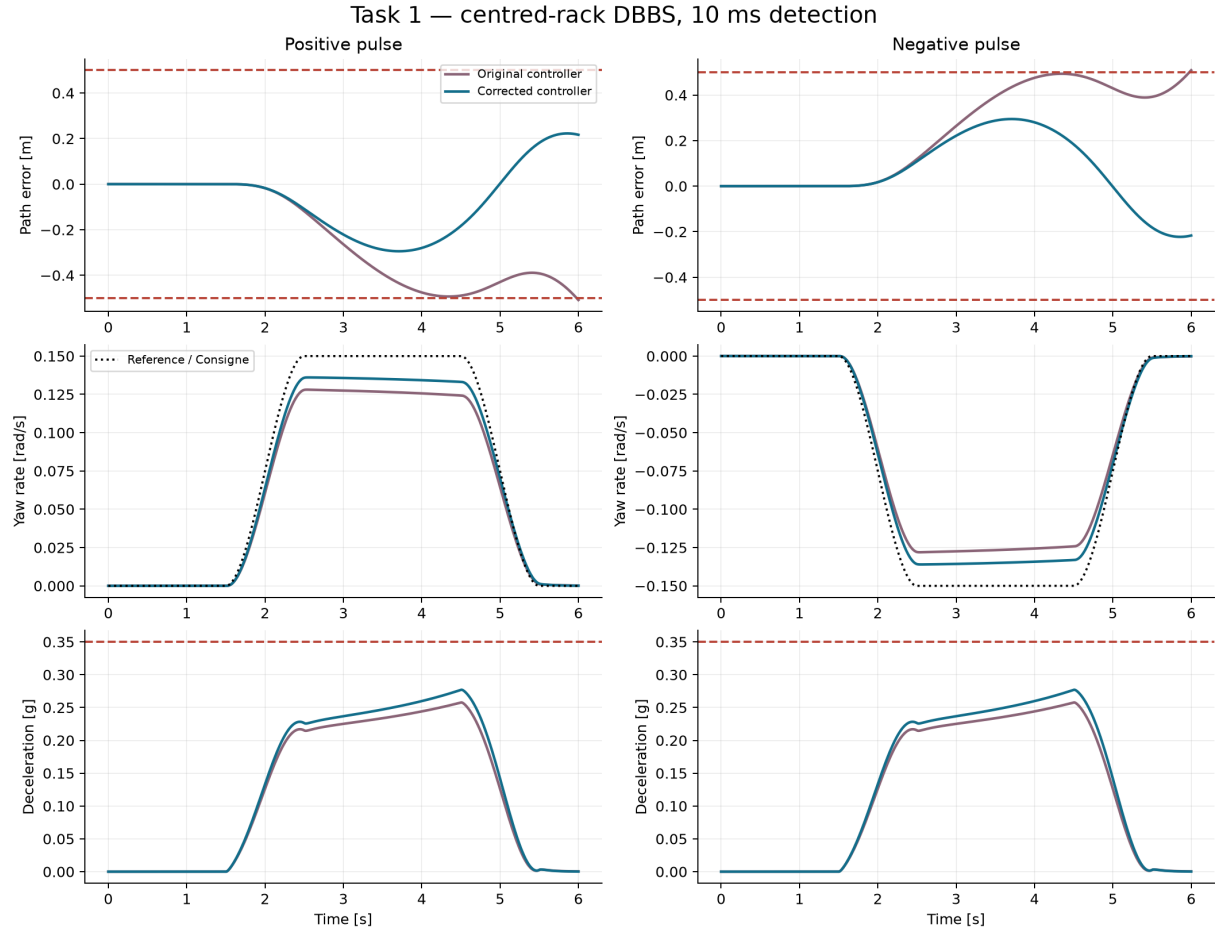


Figure 6.1 – Centred-rack DBBS before and after correction, both yaw directions, 10 ms diagnosis.

6.6 Residual braking and sideslip

The dynamic authority maneuver applies the channel loss before the heavy-braking demand. On the unchanged 1.5–1.9 s plateau, braking is normalized by μg , a nominal upper bound. This deliberately conservative denominator supports a retained-capability statement without assuming a measured maximum from a different maneuver.

Table 6.4 – Delivered braking authority in the evaluated plateau.

Condition	$a/(\mu g)$	$ \beta _{max}$, rad	$ M_z _{hold}$, N m
nominal	0.9800	0.00000	0.000
brake_fl_loss	0.6111	0.06051	10.023
brake_fr_loss	0.6111	0.06051	10.023
brake_rl_loss	0.8799	0.00723	2.663
brake_rr_loss	0.8799	0.00722	2.663

The front-loss cases retain 61.1147% braking, with peak sideslip about 0.0605 rad and hold

yaw moment about 10.0 N m. Rear-loss cases retain 87.9923%. Compared with the initial front-loss tuning, retained braking decreases by 2.5642 percentage points while peak sideslip decreases from 0.2416 to 0.0605 rad. This is a deliberate stability/capability compromise; the remaining margin above 60% is 1.1147 percentage points.

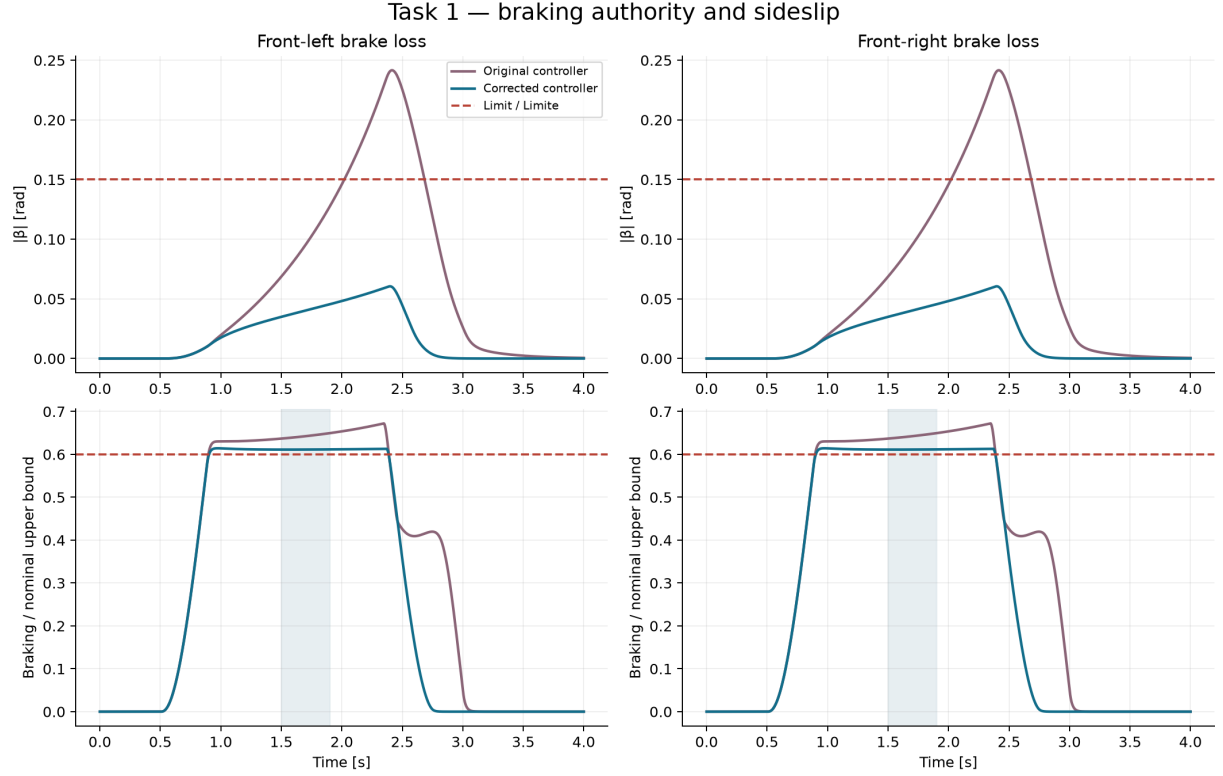


Figure 6.2 – Front-brake-loss authority: rear-grip reserve improves sideslip with a quantified braking trade-off.

6.7 Independent authority witnesses

Separate nonlinear searches construct physically feasible static witnesses independently of the allocator’s affine output. A force-aligned straight-path witness enforces zero yaw rate and moment and aligns resultant force with velocity. Its worst retained braking lower bound is 61.2636%. These witnesses establish existence at the assessed condition, not global optimality or guaranteed dynamic reachability.

$$F_{upper} = \mu mg, \quad M_{upper} = F_{upper} \max \left(\sqrt{l_f^2 + (t_w/2)^2}, \sqrt{l_r^2 + (t_w/2)^2} \right).$$

For centered DBBS at zero initial slip, the feasible yaw witness under a 0.35 g deceleration cap is approximately 4174.91 N m in either direction. Relative to the conservative nominal yaw upper bound, this gives 16.6900%, above the 15% project authority benchmark. The witness and the tracking maneuver answer different questions and are therefore reported separately.

6.8 Timing diagnostics and numerical refinement

The local audit independently recomputed all 30 original mask, affine-recovery and continuity diagnostics. The masks are applied in the diagnosis sample. All cases are already within the affine comparison band at that sample and remain there for the required dwell; the resulting zero entry delay must therefore not be interpreted as an instantaneous physical settling time. The maximum first-200-ms yaw difference is 0.013688 rad/s.

Table 6.5 – Targeted 2 ms versus 1 ms comparison.

Case	Δr_{max} , rad/s	Δv_{max} , m/s	Δe_{peak} , mm
DBBS +	0.000156	0.002192	0.727
DBBS -	0.000156	0.002106	0.731
Front-left authority	0.000589	0.002861	–
Front-right authority	0.000589	0.002860	–

The centred DBBS path result remains below 0.2953 m at 1 ms. Refinement supports numerical consistency of the selected cases. It does not replace parameter uncertainty analysis or physical vehicle testing.

6.9 Cost of diagnostic latency: oracle-gap analysis

Every fault run performs a second allocation at the same frozen state, with the same previous command and the same demand, but using the *physical* mask instead of the diagnosed one. This second solution is the oracle: the command a perfectly informed allocator would have issued at that instant. It is logged in each time series as `OracleFx_N`, `OracleMz_Nm` and the associated authority scales. Comparing the delivered nonlinear output with the oracle isolates the price of diagnostic latency from the price of the demand itself being infeasible.

$$e_y(t) = \left\| (y(t) - y_{best}(t)) \oslash y_{scale}(t) \right\|_{\infty}, \quad y = [F_x, M_z]^T.$$

We evaluate two windows. The latency window $[t_p, t_f)$ is the interval during which the controller still treats a lost effector as available. The reconfiguration window $[t_f, t_f + 100 \text{ ms}]$ follows the flag, when the command must move to its new operating point under the rate limits. The oracle remains an affine allocator prediction; the measured side of the comparison is the nonlinear plant output.

Table 6.6 – Normalized oracle gap, recomputed from the archived time series.

Case family	e_y latency 10 ms	e_y latency 50 ms	e_y reconfig.	e_y peak
Front-left brake loss	0.0241	0.0242	0.0597	0.0597
Front-right brake loss	0.0258	0.0259	0.0620	0.0620
Rear-left brake loss	0.0206	0.0207	0.0222	0.0222
Rear-right brake loss	0.0214	0.0214	0.0223	0.0223
Steering drive A loss	0.00004	0.00005	0.00005	0.0056
Steering drive B loss	0.00004	0.00005	0.00005	0.0056
DBBS, fault at 1 s	0.0000	0.0000	0.0000	0.0192
DBBS, fault at 3 s	0.0000	0.0000	0.0002	0.0138
Ramp braking, front loss	–	–	0.0898	0.0898
Ramp braking, rear loss	–	–	0.0383	0.0383

Three observations follow. First, the gap accumulated while the controller is misinformed never exceeds 0.0259 for a brake loss and is below 10^{-4} for a steering-channel loss, because the surviving drive absorbs the missing velocity contribution without a change of operating point. Second, and less obviously, the reconfiguration window is the larger contributor for every brake case: the dominant transient cost is the rate-limited move to the new command, not the delay in learning about the fault. Third, extending the imposed delay from 10 ms to 50 ms changes the peak gap by less than 10^{-3} in normalized units in every catalog case, which is the quantitative form of the insensitivity noted in section 6.5.

For the DBBS cases the peak gap does not occur at the fault at all, but during the yaw pulse itself, reaching 0.0192 for the centred rack and 0.0138 for the held cornering rack. This is consistent with the held-rack model: the fault removes no force at the instant it occurs, and the allocator only has to work when lateral demand arrives. Every value in the table stays inside the 10% normalized band of REQ-02, but the comparison target remains the affine oracle, so this supports rather than establishes the recovery clause.

6.10 Reference comparison against unconstrained allocation

Chapter 2 selects constrained quadratic allocation on the basis of published properties. To measure rather than assert the benefit, we define a clipped pseudo-inverse comparator on the identical plant, references, fault schedule and evidence pipeline. The comparator solves $u = B^\dagger(y_d - c)$ with the same affine map, then clips each component to the same physical bounds and applies the same rate limits, with no reserve policy, no trust region and no redistribution of the clipping residual. It is therefore the natural minimal alternative, and the only difference in the loop is the allocation block.

Table 6.7 – Constrained QP versus clipped pseudo-inverse on the identical scenarios.

Scenario	QP	Pseudo-inv.	Metric	Unit
DBBS-T1-D10	0.2945	[a remplir]	$ e_{lat} _{max}$	m
DBBS-T3-D10	0.2617	[a remplir]	$ e_{lat} _{max}$	m
DBBS-NEG-T1-D10	0.2945	[a remplir]	$ e_{lat} _{max}$	m
DBBS-NEG-T3-D10	0.2617	[a remplir]	$ e_{lat} _{max}$	m
Front-left authority	0.6111	[a remplir]	$a/(\mu g)$	–
Front-left authority	0.0605	[a remplir]	$ \beta _{max}$	rad
Front-right authority	0.6111	[a remplir]	$a/(\mu g)$	–
Front-right authority	0.0605	[a remplir]	$ \beta _{max}$	rad
G2-SAT	0.98	[a remplir]	saturated ratio	–

The comparison is run on the eight DBBS cases and the four authority cases, and both controllers are scored with the unchanged measurement conventions of section 6.4. Two outcomes are informative and neither is assumed in advance. If the comparator tracks the pulse comparably, the constrained formulation is justified by its bound handling and its explicit reserve policy rather than by tracking accuracy, and the report should say so. If the comparator degrades on the front-loss cases, the degradation identifies exactly which part of the QP formulation, clipping residual redistribution or reserve scheduling, produced the measured margin.

6.11 Friction-estimate mismatch

The allocator builds its bounds from $-\mu F_z$ and therefore assumes exact knowledge of both the friction coefficient and the corner loads. Every percentage in this chapter is conditioned on $\mu = 0.90$ being simultaneously the true road value and the controller’s belief. The most realistic non-actuator fault in a physical vehicle is not an effector loss but a controller that believes the road is better than it is.

Table 6.8 – Optimistic friction estimate: plant at 0.60, allocator believing 0.90.

Case	$ e_{lat} _{max}$, m	$ \beta _{max}$, rad	a_{max}/g
DBBS-T1-D10, $\mu_{plant} = \mu_{ctrl} = 0.90$	0.2945	[a remplir]	0.2771
DBBS-T1-D10, $\mu_{plant} = \mu_{ctrl} = 0.60$	[a remplir]	[a remplir]	[a remplir]
DBBS-T1-D10, $\mu_{plant} = 0.60$, $\mu_{ctrl} = 0.90$	[a remplir]	[a remplir]	[a remplir]
Front-left authority, $\mu_{plant} = \mu_{ctrl} = 0.90$	–	0.0605	0.6111 μg
Front-left authority, $\mu_{plant} = \mu_{ctrl} = 0.60$	–	[a remplir]	[a remplir]
Front-left authority, $\mu_{plant} = 0.60$, $\mu_{ctrl} = 0.90$	–	[a remplir]	[a remplir]

The matched rows separate the effect of a lower road from the effect of a wrong belief, which the mismatched rows then isolate. The sideslip domain check $|\beta| \leq 0.15$ rad and the plant-level friction-circle check remain active in all six runs, so a mismatch that drives the vehicle out of

the validity domain is reported as a domain exit rather than as a performance figure. This study converts the friction assumption from a stated limitation into a bounded one, and it is the first robustness evidence that a vehicle-specific claim would require.

6.12 Sampled envelope assessment

The earlier envelope assessment tested 300 target points at 60 frozen states. It obtained 240 feasible witnesses and 236 qualifying 50 ms endpoint constructions. Reuse with the corrected controller was checked rather than assumed: 44 single-channel sampled states have matching inputs, allowing 220 witnesses and 216 endpoint results to transfer. The 16 DBBS states changed with the controller and their earlier probe results are not automatically reused.

This sampled evidence is useful engineering information. It is not a continuous envelope proof: successful box corners do not establish a complete nonlinear reachable set, and a 50 ms endpoint does not establish a 100 ms dwell. Extending this evidence is a clearly identified future perspective.

Safety rationale and engineering synthesis

7.1 Functional safety analysis

The safety work organizes the functional architecture into braking, lateral-control and unintended-actuation concerns. FMEA links component-level causes to vehicle-level effects and compensating actions. The scope is a qualitative engineering rationale, not quantified hardware reliability or an assigned ASIL obtained from a full hazard analysis. ISO 26262-9 provides the framework for ASIL-oriented decomposition, dependent-failure and safety analyses [10].

Table 7.1 – Condensed FMEA and development perspectives.

Element / mode	Effect and current treatment	Continuation
One brake lost	Reduced/asymmetric braking; mask and redistribution	High-saturation transition dynamics
One steering drive lost	Reduced rack rate; surviving drive remains active	Road-load torque and electrical sizing
Both drives unavailable	Held rack; DBBS yaw compensation	Wider speed/friction envelope
Dedicated power / communication	Represented by equivalent isolated channel loss	Actual isolation and timeout behavior
Shared ECU / sensing	Shared dependency; ideal input assumptions	Independent monitor and estimation
Mechanical jam / stuck actuation	Different physical effect from a released channel	Dedicated models and mitigations

The complete F01–F18 analysis, including separate actuator, power and communication causes and the D01–D08 dependency register, is provided in Appendix B. The table above is its reading guide, not the full safety register.

7.2 Candidate hazard analysis and safety goals

The report refers to an ASIL D target without deriving it. We therefore record the candidate hazard classification that the target implies, so that each requirement of Chapter 3 can be attached to a safety goal rather than appearing as a free-standing number. Severity, exposure and controllability are assigned qualitatively, at the vehicle level, for a passenger car in normal road use. This is a candidate classification prepared to structure the analysis; a certified HARA requires an agreed item definition, operational situation catalog and controllability study that

this project did not have.

Table 7.2 – Candidate hazards, classification and derived safety goals.

ID	Vehicle-level hazard	S/E/C	ASIL	Safety goal
H1	Loss of requested deceleration during a braking request	S3/E4/C3	ASIL D	SG-01
H2	Unintended yaw moment during a straight-line braking request	S3/E4/C3	ASIL D	SG-02
H3	Loss of commanded lateral control	S3/E4/C3	ASIL D	SG-03
H4	Unintended or opposing rack motion	S3/E4/C3	ASIL D	SG-04
H5	Unintended brake force at one corner	S3/E4/C3	ASIL D	SG-05

Table 7.3 – Safety goals, implemented mitigation and evidence status.

ID	Safety goal	Mitigation in this demonstrator	Status
SG-01	Retain at least 60% of nominal straight-line deceleration after any single effector loss	Masked QP redistribution across the surviving corners	Tested, section 6.6
SG-02	Limit unintended yaw moment during a straight braking request to 5% of nominal yaw authority	Explicit M_z term in the QP objective with fixed priority scaling	Tested, section 6.6
SG-03	After loss of commanded steering, provide at least 15% of nominal yaw authority with bounded path deviation	DBBS with scheduled yaw feedback and held rack	Tested, sections 6.5 and 6.7
SG-04	Prevent unintended or opposing rack torque	None. The velocity-sharing model represents loss, not opposition	Not addressed; F18
SG-05	Detect and remove unintended corner brake force	None. Fault injection removes force, it never adds force	Not addressed; F17

Recording SG-04 and SG-05 with no mitigation is deliberate. Both correspond to unintended-actuation modes already listed as F17 and F18, and both require an independent monitor rather than a reallocation policy: an allocator that can only command the effectors it believes in cannot remove an effector output it did not request. This is a structural limit of the present architecture, not a tuning gap.

7.3 Fault-tolerant time interval budget

The imposed diagnosis delays of section 3.2, the 10 ms mask benchmark of section 3.3, the recovery diagnostics of section 6.8 and the oracle-gap results of section 6.9 all measure parts

of one quantity: the interval between physical fault onset and restored controlled behavior. Collecting them into a single budget makes the safety argument checkable and shows which term dominates.

$$\text{FTTI} \geq t_{\text{detect}} + t_{\text{react}} + t_{\text{settle}},$$

Table 7.4 – Budget contributions in the present demonstrator.

Contribution	Value	Basis and status
Fault detection t_d	0 / 10 / 50 ms	Imposed sensitivity cases; no sensor-based detector is implemented
Mask application	≤ 2 ms	One allocation period; sampled on the control grid
Allocation solve	[a remplir]	Measured solve time per sample against the 2 ms budget
Brake command slew, full range	≈ 106 ms front corner	4246.6 N front corner capacity at 40 kN/s; 2770.1 N rear gives ≈ 69 ms
Brake force lag	20 ms (1τ)	First-order model; 91.8% of an unsaturated step in 50 ms
Rack slew, full range	≈ 175 / 292 ms	$\pm 35^\circ$ at $400^\circ/\text{s}$ healthy pair, $240^\circ/\text{s}$ single drive
Settling within the 10% band	peak $e_y \leq 0.0898$	Measured, section 6.9; affine oracle reference

No FTTI target is assigned, because no agreed hazard analysis supplied one. The budget is nevertheless already decisive for one clause. In 50 ms the brake rate limit permits a command change of 2000 N, which is 47.1% of a front corner’s friction-limited range and 72.2% of a rear corner’s. The 50 ms recovery clause of REQ-02 is therefore physically attainable only when the required reconfiguration is smaller than those fractions; for a larger required change no allocation policy can satisfy it, and the constraint is the actuator rate, not the controller. Stating this explicitly is more useful than reporting a diagnostic band entry of zero, and it identifies the brake force-rate limit as the parameter to revisit first if the clause is retained.

7.4 Degradation states and safe state

The demonstrator switches behavior on the diagnosed mask, but it never names the resulting vehicle modes. Naming them is required before any warning and degradation concept can be written, and it clarifies what the measured results correspond to.

Table 7.5 – Modes implied by the implemented mask logic.

Mode	Entry condition	Implemented behavior
M0 nominal	All six channels healthy	Full allocation, 2% longitudinal reserve, 0.25 s yaw feedback
M1 reduced braking	One brake diagnosed lost	Redistribution; 8% rear reserve if the loss is at the front
M2 reduced steering	One drive diagnosed lost	Surviving drive takes the remaining rate demand
M3 DBBS	Both drives diagnosed lost	Rack held at its current angle; 0.15 s yaw feedback; yaw by brake asymmetry
M4 safe state	Not implemented	No minimal risk maneuver, no controlled stop, no driver handover is modeled

M4 is the honest entry. Every measured result in Chapter 6 is a transition into M1, M2 or M3 followed by continued driving; the demonstrator has no notion of terminating the degraded condition. A production concept would require the safe state, its entry criteria, the driver warning and the maximum permissible dwell in M3, none of which are within this scope.

7.5 Qualitative fault tree and independence

Let C denote an unmitigated common cause, I_S/I_B loss of command or control, J an uncontrollable rack state, A/B loss of the steering branches, D insufficient DBBS authority and R_B insufficient residual braking. A useful qualitative abstraction is

$$T_S = C \vee I_S \vee J \vee (A \wedge B \wedge D), \quad T_B = C \vee I_B \vee R_B.$$

Availability of healthy brakes alone does not guarantee sufficient backup steering; the relevant force and moment must be achievable in the current state. Similarly, two steering motors do not demonstrate independent sensing, computation, power and communication. A candidate ASIL D decomposition into B(D)+B(D) can be discussed conceptually, but actual branch independence must be justified at system level [11].

The shared dependencies identified are power and ground, communications, allocator software, state/request sensing, the common rack, road friction, diagnosis timing and physical installation. Recording them makes the architecture traceable and provides a concrete basis for future industrialization. No hardware certification is claimed by the simulation results.

7.6 Achievement of our project objectives

Table 7.6 – Summary of our contributions.

Objective	Completed contribution
Integrated modeling	13-state nonlinear plant, four brakes, two-drive common rack
Functional allocation	Constrained QP, diagnosed masks, bounded best effort
Degraded performance	Quantified catalog results, corrected DBBS and front-loss tuning
Engineering verification	Unit tests, nominal/fault campaigns, independent local audits and refinement
Traceability and reporting	Source/evidence preservation, safety rationale, requirement matrix of Appendix C, FR/EN final report

The academic premise is fulfilled through a functioning demonstrator whose retained capabilities are measured and explainable. The project does not equate this outcome with compliance with every initial all-condition requirement. The next chapter records the boundary and the technical perspectives that follow from the exploratory evidence.

Scope limits and future perspectives

8.1 Abrupt faults under fully saturated braking

Six exploratory cases introduce a front-brake loss during maximum braking, rather than before the demand. The requested deceleration is 120% of the global nominal friction bound, outside the current completion scope’s maneuver guarantees. These cases identify an actuator/contact-force modeling issue to address in a future system revision.

At the recorded onset, the healthy front command is approximately -5614.37 N and the new physical lower bound is -5123.93 N. Reaching that bound requires a 490.44 N release, while the 2 ms command-rate allowance is only 80 N. The intervals do not intersect at this state. The current implementation prioritizes the physical bound and records one rate override per exploratory case; this should not be interpreted as a physically rate-compliant actuator transition.

Table 8.1 – Exploratory transition indicators retained for future work.

Diagnosis	Min. ratio	Δr_{200ms}	Joint plateau dwell after diagnosis
0 ms	0.5025	0.05062	148 ms
10 ms	0.5059	0.06014	182 ms
50 ms	0.4959	0.10500	No complete 100 ms dwell before 1.9 s

The values are mirrored across front-left/front-right losses. The onset-inclusive yaw-moment peak is about 3919.81 N m, while peak sideslip stays below 0.0604 rad. The dwell combines braking at least 60% and the yaw plateau limit; it is descriptive and is not the independent recovery criterion. We identify these transitions as future development: the present results do not demonstrate rate compliance throughout the entire transition.

A useful continuation is to separate rate-limited actuator demand/state from instantaneous tire-contact saturation and, where justified, add suspension/load-transfer dynamics. The revised semantics would then be checked on a small targeted regression before extending the scenario set. Excluding a condition from the present report’s completion scope does not imply that the implemented controller enforces an operational exclusion automatically.

8.2 Broader envelope, recovery and robustness

The current timing result is an affine-reference diagnostic. A future independent nonlinear reachable-output calculation should distinguish mask activation, physical recovery, diagnosis latency and sustained tracking. The 15% reserve interpretation should be evaluated along tra-

jectories, with particular care for DBBS force/yaw coupling. At zero slip with a centered rack, yaw requires braking, so independent maxima do not define a rectangular feasible region.

The evaluated vehicle uses fixed mass, inertia, geometry, tire parameters and friction. Sensitivity to these assumptions, estimation errors and actuator time constants would strengthen transferability. The 1.1147 percentage-point braking margin particularly motivates uncertainty analysis before a vehicle-specific claim. A simple rule-based or pseudoinverse controller would provide an additional comparative baseline; the present before/after results concern only revisions of the QP-based system.

8.3 Industrialization perspectives

Further work can address wheel rotational dynamics and slip control, rack torque and inertia, measured motor capability, realistic sensors and diagnostic logic, independent monitoring, hardware-in-the-loop execution and real-time solver timing. Standardized maneuvers such as a double lane change or split-friction braking and a combined steering/brake-circuit loss form natural extensions. Their procedures should be specified before execution rather than relabelling the present yaw pulse as a standardized test.

General conclusion

Through this project, we deliver an integrated BbW/SbW simulation demonstrator that coordinates redundant functional actuators and adapts allocation to diagnosed channel availability. The nonlinear plant, constrained best-effort QP, fault semantics and reproducible evaluation chain form a complete academic implementation. The work connects vehicle dynamics, optimization, mechatronic architecture and safety reasoning within one coherent system.

The main quantitative outcomes are retained front-loss braking above 61%, controlled sideslip around 0.0605 rad in the authority tests, and DBBS path deviation below 0.295 m in the original maneuver catalog. The design improvements expose useful engineering trade-offs: retaining rear lateral support slightly reduces braking utilization, while faster backup yaw tracking requires more deceleration. Targeted refinement and independent saved-data checks reinforce the credibility of these findings.

We have achieved our project objective by demonstrating useful residual braking and lateral-control behavior after selected actuator losses, within the documented model and maneuver conditions. Our transparent, reusable simulation implementation provides a basis for further development. We identify a wider operating domain and industrial validation as the next technical stages, supported by the exploratory data preserved throughout our work.

Appendix A

Reproducibility and evidence organization

A.1 Reference software and source snapshot

The reference simulation uses MATLAB R2026a Update 5 and Optimization Toolbox. The main scripted campaigns do not require the legacy Simulink prototype. Controller sources executed in the final campaign are archived with the evidence, so a later worktree is not confused with the tested revision. The current report package includes a source snapshot and a SHA-256 manifest.

```
results = run_task1_validation();
```

This optional command creates a new timestamped output folder; an explicitly existing destination is rejected. It is documented for reproducibility, not executed during final report preparation. Reading results and rebuilding this report do not consume MATLAB Online simulation time.

A.2 Private evidence index

Table A.1 – Evidence hierarchy.

Baseline	Gate 2 nominal/plant/steering and initial Gate 3 campaigns
Authority studies	Nonlinear witnesses, force-aligned braking, sampled envelope
Corrected campaign	TASK1-CTRL-01 archive; SHA-256 in package manifest
Local assessment	TASK2-LOCAL-01 audit, revision 2; JSON and seven CSV tables
Final deliverable	FR/EN LaTeX sources and PDFs, source snapshot, package manifest

We verified the corrected campaign archive’s SHA-256 against the cloud copy. We preserve provenance through source matching and saved-data audits, and retain the original campaign status fields unchanged. Our public code repository excludes reports, source CDC documents and raw evidence archives.

A.3 Local checks and report rebuilding

```
python validation/test_audit_task2.py
python validation/audit_task2.py validation/results/new_audit
tectonic final_report.en.tex
tectonic final_report.fr.tex
```

The audit requires the private saved evidence at its documented locations and only reads it. The report sources use the supplied logos, project-generated plots and retained technical sources. The final output consists of two report PDFs; defense slides and scripts are intentionally not generated.

Appendix B

Full FMEA and dependency register

We retain the complete F01–F18 register established during the safety work. F01–F06 each distinguish actuator (A), dedicated power-path (P) and dedicated communication-path (C) causes. These six families therefore contain 18 causal entries; with F07–F18, the register contains 30 coded entries. This count describes the analysis, not 30 independently validated physical fault mechanisms. The tables separate the modeled consequence from proposed detection and future implementation evidence. No numerical severity, occurrence, diagnostic coverage or risk-priority number is assigned without the necessary vehicle and hardware data. “Equivalent mask” means that an isolated fail-silent loss can be represented by the same unavailable channel; it does not establish electrical or communication independence.

B.1 Brake and steering channel losses

For A entries, command/response plausibility is a candidate detection method; P entries call for local supply monitoring and isolation; C entries call for freshness, integrity and timeout monitoring. These mechanisms are proposed measures. In the tested model, diagnosis is imposed by the fault schedule.

Table B.1: Detailed channel-loss FMEA.

ID	Mode / cause	Effect / response	Evidence / continuation
F01A	Front-left brake unavailable	Reduced deceleration and asymmetric yaw; remove the channel and redistribute force.	G3-FL-D00/D10/D50, G3-BRK-FL and authority tests. Front-loss plateau retains 61.1147% braking.
F01P	Dedicated supply lost: front-left brake	Same consequence only if isolated and fail-silent.	Equivalent FL mask; physical isolation and supply diagnostics require hardware evidence.
F01C	Dedicated communication lost: front-left brake	Same consequence only if timeout produces a defined unavailable state.	Equivalent FL mask; actual network, message corruption and timeout behavior are not modeled.
F02A	Front-right brake unavailable	Reduced deceleration and asymmetric yaw; remove the channel and redistribute force.	G3-FR-D00/D10/D50, G3-BRK-FR and authority tests. Front-loss plateau retains 61.1147% braking.
F02P	Dedicated supply lost: front-right brake	Same consequence only if isolated and fail-silent.	Equivalent FR mask; physical isolation and supply diagnostics require hardware evidence.

Continued on next page

ID	Mode / cause	Effect / response	Evidence / continuation
F02C	Dedicated communication lost: front-right brake	Same consequence only if timeout produces a defined unavailable state.	Equivalent FR mask; actual network, message corruption and timeout behavior are not modeled.
F03A	Rear-left brake unavailable	Reduced deceleration and asymmetric yaw; remove the channel and redistribute force.	G3-RL-D00/D10/D50, G3-BRK-RL and authority tests. Rear-loss plateau retains 87.9923% braking.
F03P	Dedicated supply lost: rear-left brake	Same consequence only if isolated and fail-silent.	Equivalent RL mask; physical isolation and supply diagnostics require hardware evidence.
F03C	Dedicated communication lost: rear-left brake	Same consequence only if timeout produces a defined unavailable state.	Equivalent RL mask; actual network, message corruption and timeout behavior are not modeled.
F04A	Rear-right brake unavailable	Reduced deceleration and asymmetric yaw; remove the channel and redistribute force.	G3-RR-D00/D10/D50, G3-BRK-RR and authority tests. Rear-loss plateau retains 87.9923% braking.
F04P	Dedicated supply lost: rear-right brake	Same consequence only if isolated and fail-silent.	Equivalent RR mask; physical isolation and supply diagnostics require hardware evidence.
F04C	Dedicated communication lost: rear-right brake	Same consequence only if timeout produces a defined unavailable state.	Equivalent RR mask; actual network, message corruption and timeout behavior are not modeled.
F05A	Steering drive a unavailable	Reduced rack slew rate; surviving drive takes the request after diagnosis.	G3-SA-D00/D10/D50 and rack benches. Single-drive angle range is retained in the functional model; loaded torque remains to be assessed.
F05P	Dedicated supply lost: steering drive A	Same consequence only if isolated and fail-silent.	Equivalent SA mask; physical isolation and supply diagnostics require hardware evidence.
F05C	Dedicated communication lost: steering drive A	Same consequence only if timeout produces a defined unavailable state.	Equivalent SA mask; actual network, message corruption and timeout behavior are not modeled.
F06A	Steering drive b unavailable	Reduced rack slew rate; surviving drive takes the request after diagnosis.	G3-SB-D00/D10/D50 and rack benches. Single-drive angle range is retained in the functional model; loaded torque remains to be assessed.
F06P	Dedicated supply lost: steering drive B	Same consequence only if isolated and fail-silent.	Equivalent SB mask; physical isolation and supply diagnostics require hardware evidence.
F06C	Dedicated communication lost: steering drive B	Same consequence only if timeout produces a defined unavailable state.	Equivalent SB mask; actual network, message corruption and timeout behavior are not modeled.

B.2 Shared, sensing and unintended-actuation modes

Table B.2: Detailed shared and non-loss failure modes.

ID	Mode / cause	Effect / response	Evidence / continuation
F07	Both steering drives lost	Held rack; lateral control depends on DBBS authority.	Eight corrected DBBS cases meet their path/yaw/deceleration criteria. Common cause versus dual loss is not resolved by this test.
F08	Mechanical rack jam or free motion	Uncommanded steering state may exceed backup capability.	Mechanical analysis, jam/free-rack models and bounded backup envelope are future work. Held-drive loss is not a general jam test.
F09	Steering request lost, frozen or corrupt	The intended lateral motion may be unknown or incorrect.	Candidate redundant input sensing and plausibility. No request fault is represented by an actuator mask.
F10	Brake request lost, frozen or corrupt	Missing or unwanted deceleration demand.	Candidate independent request checks and defined fallback. No substitute driver intent is inferred in the model.
F11	Rack-angle measurement bias or loss	Incorrect servo error and allocation effectiveness.	Ideal angle feedback is used. Independent sensing or qualified reconstruction requires dedicated verification.
F12	Yaw or speed measurement bias or loss	Incorrect yaw feedback, tire state or operating-domain estimate.	Ideal state feedback is used. Sensor plausibility and fallback estimation remain implementation perspectives.
F13	Brake feedback fault at any corner	False isolation or delayed recognition of force loss.	Four sensor instances require per-corner supervision. Sensor dynamics and diagnostic coverage are not simulated.
F14	Allocator crash, overrun or plausible wrong output	Both force and yaw requests can be affected.	Bounded solver fallback is implemented. Watchdog, independent command monitor and qualified backup controller are separate future measures.
F15	Late, missed or false diagnosis	Unavailable actuator retained, or healthy authority removed.	0/10/50 ms imposed delay sensitivity is tested. Missed, false and intermittent flags need additional cases and detector design.
F16	Common supply, ground or communication fault	Correlated loss or corruption across domains.	Separate logical flags do not establish independence. Isolation schematics, common-cause analysis and hardware tests are required.
F17	Stuck-on or unintended brake force	Persistent asymmetric yaw or unwanted braking.	Candidate independent force supervision and disconnect path. Force-removal tests do not cover retained force.

Continued on next page

ID	Mode / cause	Effect / response	Evidence / continuation
F18	Unintended or opposing steering-drive torque	Rack motion can oppose the healthy drive or the command.	Torque supervision and disconnection require a loaded mechanical model. Velocity sharing does not establish opposing-torque tolerance.

B.3 Dependent-failure review

We use the following register to distinguish functional channel redundancy from implementation independence. These dependencies explain why a B(D)+B(D) decomposition remains a candidate system concept rather than an attribute that can be assigned to the two simulated motors alone.

Table B.3 – Shared dependencies and evidence needed for implementation.

ID	Dependency	Evidence to develop
D01	Power and ground	Supply routes, protection, isolation and common-failure analysis
D02	Communication	Bus routes, integrity, freshness and timeout behavior
D03	Shared computation	Independent monitoring, execution timing and software-fault treatment
D04	Inputs and sensors	Sensor redundancy, diagnosis and fallback estimation
D05	Common rack	Jam, free motion, road load and motor torque capability
D06	Tire adhesion	Friction/load sensitivity and simultaneous force/yaw envelope
D07	Diagnosis and timing	Physical-onset timing budget, false/missed flags and recovery
D08	Physical installation	Thermal, electromagnetic and damage-related common causes

For the qualitative fault tree, the steering top event has minimal cut sets $\{C\}$, $\{I_S\}$, $\{J\}$ and $\{A, B, D\}$; braking has $\{C\}$, $\{I_B\}$ and $\{R_B\}$. A common cause appears once and is not treated as two independent events. These logical sets identify where an implementation needs protection; they do not supply failure probabilities.

Requirements traceability

This matrix links each numerical clause of the engineering baseline to the implementation that addresses it and to the evidence that measures it. It supersedes the scattered summaries of Tables 1.2, 3.2 and 7.2 for status purposes. Four verdicts are used. *Demonstrated* means the clause was measured and met under the documented conditions. *Partial* means supporting evidence exists but does not establish the clause as written. *Not established* means the required independent evidence is absent. *Out of scope* means the clause was explicitly deferred under the revised academic scope.

Table C.1: Requirement-to-evidence matrix.

ID	Clause	Implementation and evidence	Verdict
REQ-01	Commands remain within physical, rate and mask bounds at all times	QP box constraints and rate rows; five nominal and 30 catalog cases pass the bounds, rate and friction checks	Demonstrated in the catalog
REQ-01	Same clause during abrupt faults under fully saturated braking	Six exploratory cases log a healthy-channel rate override; section 8.1	Not met; see 8.1
REQ-02.1	New actuator mask active within 10 ms	Mask applied in the diagnosis sample on the 2 ms grid; 30 re-computed diagnostics	Demonstrated as modeled
REQ-02.2	90% of post-fault best-achievable output within 50 ms, then a 10% band for 100 ms	Affine-oracle diagnostics pass; normalized oracle gap ≤ 0.0898 , section 6.9; no independent non-linear reference	Partial
REQ-02.3	Healthy commands respect 40 kN/s and 400 deg/s	Rate rows plus per-motor contribution checks in every run	Demonstrated in the catalog
REQ-02.4	Within 0.05 rad/s of nominal yaw rate for 200 ms after the fault	Maximum first-200-ms difference 0.013688 rad/s in the catalog; exceeded in the saturated exploratory cases	Demonstrated in envelope only
REQ-03a	At least 60% of nominal straight-line deceleration after any one corner loss	Minimum front-loss plateau 61.1147%; independent force-aligned static witness 61.2636%; sections 6.6 and 6.7	Demonstrated at the evaluated states
REQ-03a	Residual yaw moment $\leq 5\%$ of nominal yaw authority	Plateau hold moment 10.023 N m front, 2.663 N m rear against a 567.137 N m limit	Demonstrated at the evaluated states
REQ-03b	60% of nominal road-wheel angle range and rate after one drive loss	Functional common-rack bench; full range and 60% rate are modeling assumptions, not measured torque capability	Functional model only
REQ-03c	DBBS yaw authority $\geq 15\%$ of nominal	16.6900% conservative frozen-condition witness under the 0.35 g cap; section 6.7	Witness only, not universal

ID	Clause	Implementation and evidence	Verdict
REQ-03c	$\text{RMSE}(r) \leq 0.03 \text{ rad/s}$, $ e_{lat} \leq 0.50 \text{ m}$, $a \leq 0.35 \text{ g}$ on the pulse	8/8 cases pass; worst values 0.01146 rad/s, 0.2945 m, 0.2771 g; both centred directions retain the pass at 1 ms	Demonstrated
REQ-04	Architecture rationale, FMEA, FTA and dependent-failure analysis	F01–F18 register and D01–D08 dependencies, Appendix B; candidate HARA and FTTI budget, sections 7.2 and 7.3	Qualitative rationale only
SG-04	Prevent unintended or opposing rack torque	No mitigation; F18 requires an independent monitor	Not addressed
SG-05	Detect and remove unintended corner brake force	No mitigation; F17 requires independent force monitoring	Not addressed
Robustness	Parameter and friction-estimate robustness	Mismatch study defined in section 6.11	Out of scope
Comparison	Benefit of constrained allocation over a simpler policy	Comparator protocol defined in section 6.10	Out of scope
Extensions	Dual faults, standardized maneuvers, hardware in the loop	Structured as perspectives in Chapter 8	Out of scope

Two entries deserve emphasis because they are easy to misread as successes. REQ-02.2 is supported only by an affine-oracle comparison, and section 7.3 shows that its 50 ms clause is unattainable by construction whenever the required brake-command change exceeds roughly 47% of a front corner’s range. REQ-03b passes a bench that measures the assumptions it was built from, so it verifies implementation consistency rather than steering capability under road load.

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